

## DATA SCIENCE AND PYTHON COURSE

# Capstone Project

## Customer Churn Prediction — One-Day Build

Student: Mohammad Tofiq

### Project Overview

This capstone brings together everything you have learned so far — Python fundamentals, statistics, NumPy, Pandas, data visualization, machine learning, and feature engineering — into one complete, resume-ready project. You will build a model that predicts whether a telecom customer is likely to leave (churn), using the Telco Customer Churn dataset.

- **Dataset:** Telco Customer Churn (Kaggle)
- **Problem type:** Binary Classification (Churn: Yes / No)
- **Duration:** 4–6 hours
- **Deliverable:** One clean Jupyter Notebook with explanations, code, and a short business summary

### Session Schedule

| Time Block | Focus                          | What You'll Do                                                                                                                                                           |
|------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hour 1     | Setup & EDA                    | Load data, check shape/info/describe, find missing values, plot churn count, tenure histogram, and charges boxplot. Discuss correlation vs causation.                    |
| Hour 2     | Cleaning & Feature Engineering | Fix TotalCharges data type, handle missing values, encode categorical columns, scale numeric columns with StandardScaler, drop customerID, and split train/test (80/20). |
| Hour 3     | Model Building                 | Train a Logistic Regression baseline and a Decision Tree Classifier. Optional bonus: run K-Means on tenure and charges to explore customer segments.                     |
| Hour 4     | Evaluation & Wrap-Up           | Compare models using accuracy, precision, recall, and confusion matrix. Decide which metric matters most for churn. Write 3–4 business insight bullets.                  |

### Skills You Are Applying

Every step of this project connects directly to a topic from the course:

- **Python & Statistics:** Pandas for loading and exploring data; descriptive statistics and spread of data for summarizing churn patterns.
- **Visualization:** Matplotlib and Seaborn charts to understand distributions and relationships.
- **Feature Engineering:** Encoding categorical data and scaling numeric features with StandardScaler.
- **Machine Learning:** Train/test split, Logistic Regression, Decision Tree Classifier, and model evaluation metrics.
- **Bonus:** K-Means Clustering to view customer segments beyond the prediction task.

## What to Submit

---

Submit a single, well-organized Jupyter Notebook containing:

- Clear markdown headers for each of the four session sections
- Code cells with short explanatory comments
- Charts that support your findings
- A 4–5 line summary at the end covering: the problem, your approach, the best-performing model, and one key business insight

## Key Reminder

---

**Think like a business analyst, not just a coder.**

A good accuracy score is not the finish line. Ask yourself: which mistake costs the company more — telling a loyal customer they might leave, or missing a customer who actually leaves? This is what interviewers want to hear you explain.

---

*Good luck — build something you're proud to show in an interview!*